



Scott Cunningham

built the literature

Scott Cunningham · Baylor University

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Causal Inference: The Remix · Yale University Press · forthcoming
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The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including
disadvantaged adult men.

Random assignment to treatment vs. control. Outcome: real earnings in 1978.

LaLonde's experimental benchmark

$$\widehat{ATT}^{\text{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

Random assignment $\Rightarrow$ no selection. This is the truth.

Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the
CPS and the **PSID**.

*A simulation of what an applied researcher would face —
no experiment, just observational data and methods.*

Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

Could any of them recover \$886?

No

Estimates ranged from $-\$16,000$ to $+\$3,000$.

Most were negative. Most missed the benchmark by orders of magnitude.

LaLonde (1986), AER 76(4): 604–620.

The headline that shook applied econometrics

*“The econometric procedures
cannot replicate the experimental results.”*

*This is the paper that made program evaluation
move toward randomization and design.*

Dehejia & Wahba reopened the case

Rajeev Dehejia and **Sadek Wahba**,
Review of Economics and Statistics (2002).

They asked: does **propensity score matching**
do better than LaLonde's methods?

First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in
both 1974 and 1975.

A smaller, slightly different population.

This is the subsample the homework uses: `lalonge_exp_panel.dta`.

A new experimental benchmark

$$\widehat{ATT}_{D\&W}^{RCT} \approx \$1,794$$

Roughly **\$1,700** in the way we usually report it.

Why bigger than LaLonde's \$886? Different subsample, different ATT.

Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

This is not a contradiction — it is the ATT defined on different units.

D&W's quasi-experimental result

With **propensity-score matching** on
CPS / PSID controls...

they recovered $\approx$ \$1,700.

A constructive answer to LaLonde: design plus matching can work.

A puzzle

We used **non-experimental controls** (CPS, PSID)
in the larger LaLonde–D&W literature.

And yet the ATT we target is still $\approx \$1,700$.

Why doesn't the ATT change when we change the controls?

Because the ATT is a property of the treated

$$ATT = \mathbb{E}[Y^1 - Y^0 \mid D = 1, \text{Post}]$$

Only the $D = 1$ side appears.

The control units are just a *tool* to learn about $\mathbb{E}[Y^0 \mid D=1, \text{Post}]$.

Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D=1] = \mathbb{E}[Y^0 \mid D=0]$$

So $\mathbb{E}[Y \mid D=0, \text{Post}]$ *is* the counterfactual we need.

No selection bias.

Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D=1] \neq \mathbb{E}[Y^0 \mid D=0]$$

Selection bias enters.

NSW men had worse pre-program earnings. The CPS pool didn't.

But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D=1, \text{Post}]$ is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —
and the size of the bias.

So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at \$1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

Did it?

The benchmark and the challenge

RCT benchmark: \$1,794

Naive DiD on the non-experimental panel: \$3,621

We can do better than additive controls. Let's go to the code.

Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of
American households from the **CPS**.

Data: lalonde_nonexp_panel.dta. Same treated rows. New controls.

Selection is severe

	1974	1975	1978
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15,992$)	\$14,017	\$13,651	\$14,847
Diff	-\$11,921	-\$12,119	-\$8,498

$$\mathbb{E}[Y^0 \mid D=1] \ll \mathbb{E}[Y^0 \mid D=0]$$

The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.

The 2×2: where \$3,621 comes from

	1974	1975 (pre)	1978 (post)
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15,992$)	\$14,017	\$13,651	\$14,847

NSW change: $\$6,349 - \$1,532 = \$4,817$

CPS change: $\$14,847 - \$13,651 = \$1,196$

DiD $\$4,817 - \$1,196 = \$3,621$

$\$3,621 \neq \$1,794$

NSW men bounce back fast from a deep dip. CPS men don't. DiD attributes the entire gap in growth rates to the program.

Naive DiD on this sample

$$\hat{\delta}_{\text{naive DiD}} = +\$3,621$$

Right sign. Wrong magnitude. The benchmark is \$1,794.

*NSW men had unusually low pre-earnings (\$1,532) and bounced back fast.
CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.*



Spec 0: Unconditional parallel trends, no covariates

Assumption. Naive parallel trends: $E[\Delta Y(0) \mid D=1] = E[\Delta Y(0) \mid D=0]$ unconditionally.

Run:

```
reg re i.post##i.ever_treated, robust
```

Compare to \$1,794.

Spec A: Conditional PT, additive X

Assumption. Conditional parallel trends with one slope per covariate (pre = post).

Run:

```
global X "age_agesq_educ_educsq_marr_nodgree_black_hisp_re74_re75_u  
reg re i.post##i.ever_treated $X, robust
```

Compare to \$1,794.

Spec B: Conditional PT, Post $\times X$

Assumption. Conditional parallel trends, allowing X -effects to differ pre vs post. Use ## so the spec includes *both* the pre-period X effect and its post-period change.

Run:

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##c.re75 i.post##i.u74 i.post##i.u75, robust
```

Compare to \$1,794. *Single-# forces pre-period X effect to zero — not what we want.*

Spec C: Saturated TWFE / OR (HIT 1997)

Assumption. Conditional parallel trends; impute $Y(0)$'s trend on controls only and average over treated.

Run — by hand:

```
reshape wide re, i(id) j(post)
gen dy = re1 - re0

reg dy $X if ever_treated == 0
predict dy_hat
gen delta_hat = dy - dy_hat if ever_treated == 1
summarize delta_hat if ever_treated == 1
```

Cross-check with drdid:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) reg
```

Compare to \$1,794. *Two numbers should agree to the dollar.*

Spec D: IPW (Abadie 2005)

Assumption. Conditional PT; correct propensity score. Treated weighted by 1; controls weighted by $\hat{p}(X)/(1 - \hat{p}(X))$.

Run — by hand:

```
logit ever_treated $X
predict phat, pr
quietly sum ever_treated
local pr_treat = r(mean)
gen w_ipw = (ever_treated - phat) / (1 - phat) / 'pr_treat'
gen ipw_dy = w_ipw * dy
summarize ipw_dy
```

Cross-check with drdid:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

Compare to \$1,794.

Spec E: DR-DiD (Sant'Anna & Zhao 2020)

Assumption. Conditional PT, AND *either* $\hat{\mu}_0$ or $\hat{p}(X)$ is correct (not both required).

Run — by hand (in wide-FD form, phat/dy_hat already computed):

```
gen dr_t = ever_treated * (dy - dy_hat) / 'pr_treat'  
gen dr_c = (1 - ever_treated) * (phat / (1 - phat)) * (dy - dy_hat) / 'pr_treat'  
quietly sum dr_t  
local mt = r(mean)  
quietly sum dr_c  
display "DR_ATT=" %9.0fc 'mt' - r(mean)
```

Cross-check with drdid:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Compare to \$1,794.

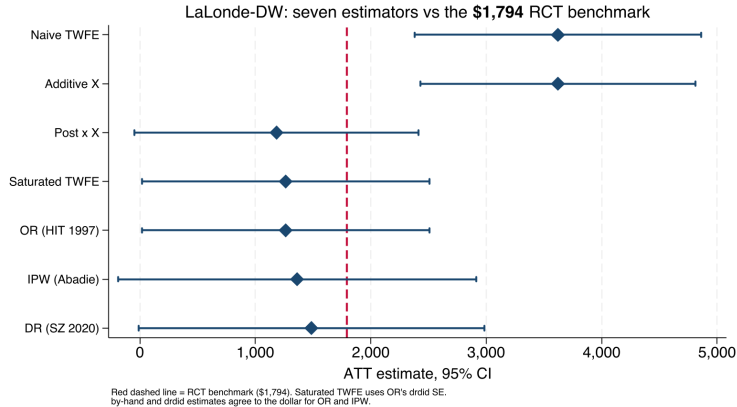
each one get?

The comparison

Spec	ATT	Distance from \$1,794
0. Naive TWFE (no X)	+\$3,621	+\$1,827
A. Additive X	+\$3,621	+\$1,827
B. Post $\times X$ (## syntax)	+\$1,184	-\$610
C. Saturated TWFE (FD)	+\$1,263	-\$531
OR by hand = <code>drdid reg</code>	+\$1,263	-\$531
D. IPW by hand = <code>drdid ipw</code>	+\$1,361	-\$433
E. DR by hand	+\$1,496	-\$298
DR <code>drdid dripw</code>	+\$1,485	-\$309
RCT benchmark	+\$1,794	—

Naive = Additive- X : the linear X control is absorbed. DR closest at \$298 below. By-hand = `drdid` to the dollar for OR and IPW; DR within \$11.

Seven specs, one benchmark



Naive and Additive-X CIs sit entirely above \$1,794. Every design-aware spec contains it.

DR lands closest; the others bracket it

- **DR** (\$1,496) sits **\$298 below** — closest observational estimator.
- **IPW** (\$1,361) sits \$433 below.
- **OR / Saturated TWFE** (\$1,263) sits \$531 below — two paths, same number.
- **Post** $\times X$ (\$1,184) sits \$610 below — still misses Bias_{HTE} .
- **Additive** X (\$3,621) doesn't move off Naive (\$3,621) — the FE absorbed the linear X .

“Doubly robust” = robust to one misspecification, not best in every sample. DR’s 298 below is selection on unobservables — the residual no observational X can close.

What this lab teaches you

- The **ATT is fixed** ($\approx +\$1,794$). The controls only change the bias.
- **Additive X barely helps:** \$3,621 vs naive \$3,621. You need a design-aware estimator.
- $\text{Post} \times X$, OR, IPW, DR offer four different uses of the same X , all closer to truth.
- On this dataset, **DR-DiD lands closest** at \$298 below \$1,794.
- No method is magic. Each rests on an assumption you can name.

What additive X actually assumed

The additive- X spec $Y_{it} = \alpha + \delta(D_i \cdot \text{post}_t) + X_i' \beta + \varepsilon$ was a **TWFE specification choice**. Two restrictions came along free of charge:

1. **Homogeneous treatment effect in X .** One scalar δ . No room for $\delta(X)$.
2. **No time-varying X effect.** One β , used in both pre and post. If X 's relationship with $Y(0)$ shifts between periods, the spec can't see it.

Neither restriction is identification. Both are specification.

Two TWFE relaxations

We saw both in the synthetic simulation. Both are still TWFE — just less restrictive parameterizations.

Post $\times X$. Let β differ pre vs post:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \varepsilon$$

Relaxes restriction (2). Still imposes (1): one scalar δ .

Fully saturated. Let β differ pre/post *and* let δ depend on X :

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \beta_3 XD + \beta_4 DTX + \varepsilon$$

Relaxes both. The ATT is now $\delta + \beta_4 \mathbb{E}[X \mid D=1]$ — average the X -specific effect over the treated.

The punchline

Additive X failed on LaLonde
not because the data are weird,
but because the **TWFE specification** was too restrictive.

Fully saturated TWFE = the regression form of HIT 1997 outcome regression.
Same estimator. Same ATT. Different machinery.

Spec C (FD with $D \times X$ interactions) and OR-by-hand returned the same dollar: \$1,263.

whether your method
can **see it**.
